



P19

The Structural-Change Review Gate: Forcing Test Assertions in a Versioned Repository

结构性变更审核门禁：测试断言强制

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The Structural-Change Review Gate: Forcing Test Assertions in a Versioned Repository

Working Title (EN): The Structural-Change Review Gate: Forcing Independent Test Assertions on Every MatrixConfig Commit, Extending UDOS P12 with the Ouroboros Lesson

In-volume engineering paper P19 • upgrading v7.5.0 P12 "Evidence-grading-driven reproducible systems engineering" from "post-hoc grading" to "pre-commit gate" • using Ouroboros's reviewed commits as reference • using P12's "version number incident" as counterexample • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: UDOS already has a versioned repository and CHANGELOG, but "structure changed" long relied on people remembering to write tests. Ouroboros (arXiv:2608.08311) proves: **making "reviewed commits" the runtime base for later runtimes, self-development only then continues without collapsing.** This paper lands it as a hard UDOS gate—any MatrixConfig parameter change must be accompanied by independent test assertions to merge; and clearly this is a **design proposal • to implement.**

Evidence caliber (stated only once in the whole paper): this paper is an engineering governance design paper, **containing no new UDOS measured numbers.** The external paper Ouroboros (arXiv:2608.08311, existence verified online 2026-09-21)'s Terminal-Bench 2.1 number (original 86.97%, after removing one reward-hack 86.74%) is **self-reported SOTA**, not independently reviewed; this paper does not cite that score as evidence for UDOS gate effectiveness, only borrowing its "reviewed commit → runtime" engineering structure. UDOS old events (P12 version number incident) point back to v7.5.0 P12. This paper's gate is a **design proposal • to implement**, benefits without experiment support marked [RESULT NEEDED] .

Abstract

For a self-improving Agent system, the most dangerous failure is not "changed badly" but "**quietly changed badly, CHANGELOG still recording it as improvement.**" UDOS v7.5.0's P12 paper established evidence grading (verified / cpu-proto / cpu-proxy / unverified), but it is **post-hoc**: after changes already merged and run, grading then decides how hard the sentence can be stated. This paper fills the missing **pre-hoc** link of P12—the structural-change review gate. Triggering it is a concrete lesson: P12 itself recorded a "version number incident"—some commit changed the version number prefix convention, without any test assertion catching it, causing downstream parsing versions by the old convention to silently fail (event description pointing back to P12, this paper not re-characterizing). Ouroboros (arXiv: 2608.08311, Ouroboros: A Self-Developing Frontier Coding Agent with Reviewed Core Evolution) gives the positive reference: its tools, prompts, context assembly and core implementation all continuously improve through **reviewed commits**, and these commits become the runtime for later work; the common point of its two evolution modes (recursive free evolution / experience-driven core evolution) is—every structural change must pass a review. This paper lands this structure as UDOS's gate specification: dividing changes into two classes—**behavioral changes** (not changing MatrixConfig structure, going through normal CI) and **structural changes** (changing MatrixConfig parameter schema, topology defaults, contract fields, stopping decision thresholds); the latter before merging must simultaneously satisfy three—(i) at least one new independent test assertion directly covering the changed structural quantity; (ii) key regression test sets all green; (iii) CHANGELOG entries marking

change class and evidence grade. This paper also points out Ouroboros's drawback: its "review" relies on human reviewers, review bandwidth being the hard upper limit of self-improvement speed; the UDOS gate therefore does not pursue "automatically approving everything" but the mechanical, falsifiable, not-relying-on-people-remembering rule of "structural change lacking assertions is rejected from merging." This paper is a design proposal, effects pending [RESULT NEEDED].

Keywords: review gate; structural change; test-driven; versioned repository; evidence grading; CI/CD; self-improvement safety; UDOS P12; Ouroboros

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** P12's evidence grading is post-hoc; structural changes if not forced to accompany test assertions will repeat the "version number incident"—changed structure, CHANGELOG recording as improvement, downstream silent mismatch.
 - **Core argument:** singling out "structural change" from ordinary code changes, before merging forcing the three-piece set—**independent test assertions + regression all green + CHANGELOG classified marking**; lacking assertions mechanically rejected, not relying on people remembering.
 - **Evidence clues:** (1) P12 version number incident (pointing back to P12's counterexample); (2) Ouroboros's reviewed-commits structure (arXiv:2608.08311, existence verified, score self-reported); (3) P2/P10/P11's existing structural quantity test basis (24/24 stopping decision acceptance, 26 TransferBundle contract tests, internal market conservation, pointing back to papers).
 - **Contribution:** defining UDOS's "structural change" boundary; giving the three-condition gate specification; pointing out Ouroboros review bandwidth's drawback and designing the gate as a mechanical rule not relying on human bandwidth; giving falsifiable criteria.
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1 Problem: Post-hoc Grading Cannot Save Pre-hoc Silence

1.1 What P12 Did Right and Did Not

P12 paper's core contribution is **attaching evidence grades to every sentence**. A sentence saying "24/24 acceptance correct," it marks verified; saying "8 Agent throughput hump," it marks cpu-**proto**. This grading lets readers know how far to trust. But it has an implicit premise: **the change itself is cleanly, reproducibly recorded**. If one structural change quietly changes the version number prefix, or changes a TransferBundle field's semantics without changing tests, then post-hoc however graded, what comes out is "truth based on wrong configuration"—however fine the evidence grade, cannot recover a silently polluted baseline.

1.2 Counterexample: The Version Number Incident

P12 recorded one such incident (this paper only citing its lesson, not restating details): one commit changed the version number prefix convention, the repository had no test assertion "version number must match that prefix regex." As a result downstream parsing by the old convention, the new-format version number wrongly classified, while CI was green—because nothing was checking this. This is precisely the "post-hoc grading" blind zone: the grading system's input (current baseline) itself polluted, however strict grading also scoring the polluted baseline.

1.3 Positive Reference: Ouroboros's reviewed commits

Ouroboros (arXiv:2608.08311) is a self-developing frontier coding Agent, its core mechanism being: **tools, prompts, context assembly, core implementation all continuously improve through reviewed commits, and these commits become the runtime for later work.** It has two evolution modes—recursive free evolution (improvement itself a task, after completion scheduling the next round) and experience-driven core evolution (daily work exposing bugs and inefficiency, producing reviewed structural changes). The paper self-reports Terminal-Bench 2.1 reaching 86.74% (original 86.97%, after removing one reward-hack), the benchmark's reported SOTA (self-reported, not independently reviewed).

This paper does not borrow the 86.74 number. This paper borrows the structure: **every change making later runtimes change must pass a review, and the reviewed result explicitly recorded as part of the runtime.**

Figure 1 Post-hoc evidence grading vs pre-hoc review gate

Figure 1 Left: P12 post-hoc grading—changes first merged, run then grades attached; right: this paper's pre-hoc gate—structural changes before merging forced independent assertions. Conceptual illustration, not measured data.

2 Gate Specification (design proposal • to implement)

2.1 What Counts as "Structural Change"

First draw the boundary, not all commits go through the gate. This paper divides changes into two classes:

- **Behavioral changes:** not changing MatrixConfig's schema, not changing topology defaults, not changing TransferBundle contract fields, not changing BFT-lite stopping thresholds, not changing settlement conservation rules. These go through normal CI.
- **Structural changes:** touching any of the above. Examples include—adding/removing MatrixConfig parameters, changing their defaults or units, changing P1 topology decision tree branch conditions, changing the $\$f\$$ definition in P2's quorum $\$2f+1\$$, changing P10 TransferBundle field semantics, changing P11 settlement accounting units.

The decision rule itself written as a mechanical check: whether the diff touches the three directories `matrix_config/`, `contracts/`, `consensus/` or corresponding schema files. Touching triggers the gate, not relying on human subjective judgment "does this count as structural change."

2.2 The Three-Condition Gate

One structural change commit, before merging must simultaneously satisfy:

1. **Independent test assertions:** at least add one test, directly asserting the changed structural quantity. "Independent" means it is not the changed code's internal unit self-test but an assertion written from an external contract perspective—e.g. changed the version number prefix, must have one test asserting `re.match(VERSION_RE, version)` not empty; changed a TransferBundle field, must have one contract test asserting handoff rejected when that field missing.
2. **Regression all green:** P2 stopping decision acceptance set, P10 TransferBundle contract test set, P11 settlement conservation test set all pass (these test sets already exist, pointing back to papers, verified tests guarding).

3. **CHANGELOG classified marking:** CHANGELOG adds one entry, marking [structural] class, corresponding paper number, evidence grade (e.g. cpu-proto). Missing marking = missing gate.

Three lacking any one **mechanically rejected from merging**. This rule does not rely on the reviewer remembering to ask "did you write tests"—CI directly blocks.

2.3 Relationship with Existing Structural Quantity Tests

UDOS does not start from zero. P2's BFT-lite stopping decisions have acceptance tests (committee 24/24 acceptance correct, verified); P10's TransferBundle has 26 contract tests (verified); P11's internal market settlement conservation guaranteed by code structure. The gate does **elevating these existing tests from "post-hoc regression" to "precondition for structural change"**: before "after changing code conveniently running once," now "no new assertions not allowed to change structure."

Table 1 Gate contrast of behavioral change vs structural change (design specification)

Dimension	Behavioral change	Structural change
Touched scope	not changing schema/ contract/threshold	changing MatrixConfig schema, topology defaults, TransferBundle fields, stopping thresholds, conservation rules
New assertions	not forced	forced at least one independent contract assertion
Regression requirement	normal CI	P2/P10/P11 all test sets green
CHANGELOG	normal entry	must [structural] + paper number + evidence grade
Lacking assertions	normally merged	mechanically rejected from merging
Evidence grade	design proposal • to implement	design proposal • to implement

Figure 2 Structural change merge pipeline

Figure 2 Structural change commit → mechanical decision whether touching structural directories → three-condition gate → if passed merged and becomes runtime; any condition missing rejected. Conceptual illustration.

3 Drawbacks / Failure Boundaries / Falsifiable Conditions

3.1 Drawbacks (restraint that must be kept when citing Ouroboros)

1. **Review bandwidth is the hard upper limit.** Ouroboros's "reviewed commit" path relies on reviewers; if reviewers are human, then self-improvement's recursive depth is **capped by human bandwidth**—this is not true autonomous evolution but "rapid iteration under human review." It modifies the Harness, not the review rule "determining later modifications itself" (consistent with P14's L2/L5 distinction).
2. **Self-reported score does not equal gate effectiveness.** Terminal-Bench 86.74% is self-reported SOTA, cannot infer "review gate improved UDOS quality." This paper borrows structure, not score.
3. **Gate too strict suppresses evolution.** If every structural change requires one new contract assertion, evolution speed may be slowed by test-writing cost—precisely why "behavioral changes" need exemption, otherwise the gate becomes shackles.

3.2 Failure Boundaries

- If the gate only checks "whether new tests added" without checking "whether tests really cover the changed structure," it will be bypassed by "adding one irrelevant assertion." This requires independent assertion review still needing human/AI one look at assertion-diff relevance—mechanical rules cannot block this, the gate's inherent gap, this paper not pretending to solve.
- If the structural change itself changes the "what counts as structural change" decision rule (self-reference), mechanical directory checks fail—such meta-changes must be explicitly marked and go human, cannot automatically pass the gate.

3.3 Falsifiable Conditions

- **F19-a:** if in one iteration cycle after introducing the gate, "version number incident"-class silent structural failures (changed structure yet no corresponding assertion) occurrence count did not decrease relative to before introduction, then the gate exists in name only, needing redesign of trigger rules.
- **F19-b:** if the gate makes structural change merge throughput drop beyond the design tolerance threshold (concrete threshold pending [RESULT NEEDED] calibration), then need relaxing the "behavioral/structural" boundary, otherwise the gate will kill P14-style rapid iteration.
- The above are post-hoc-scorable criteria under a design proposal, not established conclusions.

4 Conclusion

P12 taught UDOS "how hard to state a sentence," P19 wants to teach UDOS "changed structure must leave evidence." Ouroboros's reviewed commits proved this path sustainable, but its drawback reminds us: **self-improvement relying on human review bandwidth is not true autonomy**. UDOS's countermeasure is mechanizing the most critical step—structural change lacking independent test assertions rejected from merging—letting this rule not rely on anyone remembering. It does not pursue automatically approving everything, only not letting "changed badly yet recorded as improvement" silently occur.

Post-hoc grading determines how long a sentence can be trusted, pre-hoc gate determines whether one change can quietly poison the baseline. The version number incident already paid tuition once, the gate is this tuition's receipt.

References

Software engineering and testing foundations

1. Beck, K. (2002). Test-Driven Development: By Example. Addison-Wesley. (classic test-first methodology)
2. Martin, R. C. (2008). Clean Code: A Handbook of Agile Software Craftsmanship. Prentice Hall. (design by contract and boundaries)

2026 frontier material (existence verified online 2026-09-21; performance numbers self-reported, not independently reviewed)

1. Ouroboros (2026). Ouroboros: A Self-Developing Frontier Coding Agent with Reviewed Core Evolution (v3). arXiv:2608.08311. (reviewed commits become runtime; Terminal-Bench 2.1 self-reported 86.74%)

UDOS paper volumes (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P2 "BFT-lite safety verification of multi-agent stopping decisions" (24/24 acceptance, verified), P10 "TransferBundle handoff information loss measurability" (26 contract tests, verified), P11 "Internal task market settlement conservation," P12 "Evidence-grading-driven reproducible systems engineering" (version number incident counterexample), P14 "RSI convergence" (L2 changing Harness / L5 structural self-modification).
 2. UDOS v7.6.0 paper volume: P33 "v7.6.0 architecture master: eight structural change specifications and compatibility paths" (review gate registered as structural change item 4).
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Evidence Discipline and Reproduction Notes

- This paper is an **engineering governance design paper**, no new experiments; three-condition gate, [structural] marking rules, F19-a/b criteria are all design proposals, merge rate and incident count improvement [RESULT NEEDED], forbidden to write as measured benefits.
- Ouroboros's Terminal-Bench 86.97%/86.74% is self-reported SOTA, not independently reviewed, only structural reference.
- P12 version number incident is a v7.5.0 recorded event, this paper only citing its lesson, not re-characterizing details.
- The gate's human relevance review of "whether assertions really cover the changed structure" is an inherent gap, this paper honestly marking, not claiming automated closure.